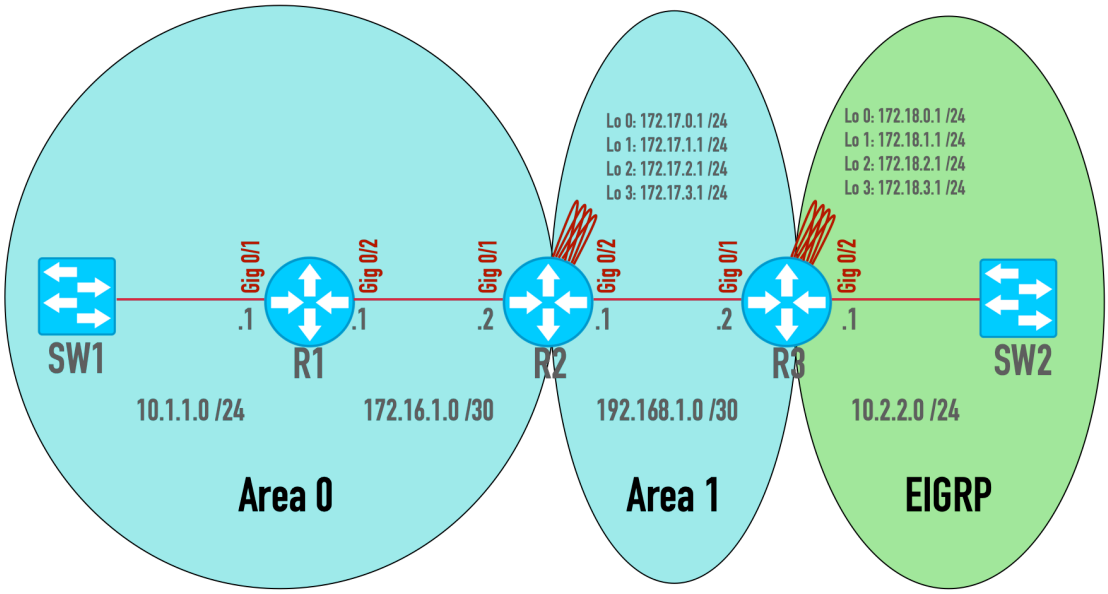
OSPF Route Summarization

Topology



Initial Configuration Commands

R1:

```
enable
conf t
host R1
no banner motd
no banner login
no banner exec
no banner incoming
line vty 0 15
password cisco
login
exec-timeout 0 0
line con 0
logging synchronous
exit
no ip-domain lookup
int gig 0/1
ip address 10.1.1.1 255.255.255.0
no shut
int gig 0/2
ip address 172.16.1.1 255.255.255.252
```

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```
no shut
router ospf 1
router-id 1.1.1.1
network 0.0.0.0 255.255.255.255 area 0
end
copy run star
```

R2:

```
enable
conf t
host R2
no banner motd
no banner login
no banner exec
no banner incoming
line vty 0 15
password cisco
login
exec-timeout 0 0
line con 0
logging synchronous
exit
no ip-domain lookup
int loopback0
ip address 172.17.0.1 255.255.255.0
int loopback1
ip address 172.17.1.1 255.255.255.0
int loopback2
ip address 172.17.2.1 255.255.255.0
int loopback3
ip address 172.17.3.1 255.255.255.0
int gig 0/1
ip address 172.16.1.2 255.255.255.252
no shut
int gig 0/2
ip address 192.168.1.1 255.255.255.252
no shut
router ospf 1
router-id 2.2.2.2
network 172.16.1.0 0.0.0.3 area 0
network 172.17.0.0 0.0.255.255 area 1
network 192.168.1.0 0.0.0.3 area 1
end
copy run star
```

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R3:

```
enable
conf t
host R3
no banner motd
no banner login
no banner exec
no banner incoming
line vty 0 15
password cisco
login
exec-timeout 0 0
line con 0
logging synchronous
exit
no ip-domain lookup
int loopback0
ip address 172.18.0.1 255.255.255.0
int loopback1
ip address 172.18.1.1 255.255.255.0
int loopback2
ip address 172.18.2.1 255.255.255.0
int loopback3
ip address 172.18.3.1 255.255.255.0
int gig 0/1
ip address 192.168.1.2 255.255.255.252
no shut
int gig 0/2
ip address 10.2.2.1 255.255.255.0
no shut
router eigrp 1
network 10.2.2.0 0.0.0.255
network 172.18.0.0
redistribute ospf 1 metric 1000000 2 255 1 1500
router ospf 1
network 192.168.1.0 0.0.0.255 area 1
router-id 3.3.3.3
redistribute eigrp 1 subnets
end
copy run star
```

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Lab Tasks

1. On R1, let's see what networks we know about right now.
2. On R3, let's summarize the 172.18 networks assigned to R3's loopback interfaces.
3. On R1, let's see if we can find our summarized network from R3 by using the "show ip route" command.
4. On R2, let's summarize the 172.17 networks assigned to R2's loopback interface. The 172.17 networks will be redistributed from OSPF Area 1 into OSPf Area 0.
5. Let's go back to R1 and see if we can now see the summarized network from R2 in R1's routing table.

Solution

Step 1: On R1, let's see what networks we know about right now.

```
R1>en
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static
route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
C        10.1.1.0/24 is directly connected, GigabitEthernet0/1
L        10.1.1.1/32 is directly connected, GigabitEthernet0/1
O E2     10.2.2.0/24 [110/20] via 172.16.1.2, 00:01:08, GigabitEthernet0/2
      172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C        172.16.1.0/30 is directly connected, GigabitEthernet0/2
L        172.16.1.1/32 is directly connected, GigabitEthernet0/2
      172.17.0.0/32 is subnetted, 4 subnets
O IA     172.17.0.1 [110/2] via 172.16.1.2, 00:01:30, GigabitEthernet0/2
O IA     172.17.1.1 [110/2] via 172.16.1.2, 00:01:30, GigabitEthernet0/2
O IA     172.17.2.1 [110/2] via 172.16.1.2, 00:01:30, GigabitEthernet0/2
O IA     172.17.3.1 [110/2] via 172.16.1.2, 00:01:30, GigabitEthernet0/2
      172.18.0.0/24 is subnetted, 4 subnets
O E2     172.18.0.0 [110/20] via 172.16.1.2, 00:01:08, GigabitEthernet0/2
O E2     172.18.1.0 [110/20] via 172.16.1.2, 00:01:08, GigabitEthernet0/2
O E2     172.18.2.0 [110/20] via 172.16.1.2, 00:01:08, GigabitEthernet0/2
O E2     172.18.3.0 [110/20] via 172.16.1.2, 00:01:08, GigabitEthernet0/2
      192.168.1.0/30 is subnetted, 1 subnets
O IA     192.168.1.0 [110/2] via 172.16.1.2, 00:01:30, GigabitEthernet0/2
```

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(##Notice how we know about the individual 172.17 routes and 172.18 routes. This tells us that nothing has been summarized yet.)
(##Doing the binary math we can see that we have the first 22 bits in common among all of those networks. This means that our subnet mask will be a /22.)

Step 2: On R3, let's summarize the 172.18 networks assigned to R3's loopback interfaces.

```
R3>en
R3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#router ospf 1
R3(config-router)#summary-address 172.18.0.0 255.255.252.0
R3(config-router)#end
```

Step 3: On R1, let's see if we can find our summarized network from R3 by using the "show ip route" command.

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static
route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
C        10.1.1.0/24 is directly connected, GigabitEthernet0/1
L        10.1.1.1/32 is directly connected, GigabitEthernet0/1
O E2     10.2.2.0/24 [110/20] via 172.16.1.2, 00:15:59, GigabitEthernet0/2
      172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C        172.16.1.0/30 is directly connected, GigabitEthernet0/2
L        172.16.1.1/32 is directly connected, GigabitEthernet0/2
      172.17.0.0/32 is subnetted, 4 subnets
O IA     172.17.0.1 [110/2] via 172.16.1.2, 00:16:21, GigabitEthernet0/2
O IA     172.17.1.1 [110/2] via 172.16.1.2, 00:16:21, GigabitEthernet0/2
O IA     172.17.2.1 [110/2] via 172.16.1.2, 00:16:21, GigabitEthernet0/2
O IA     172.17.3.1 [110/2] via 172.16.1.2, 00:16:21, GigabitEthernet0/2
      172.18.0.0/22 is subnetted, 1 subnets
O E2     172.18.0.0 [110/20] via 172.16.1.2, 00:02:29, GigabitEthernet0/2
      192.168.1.0/30 is subnetted, 1 subnets
```

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```
O IA     192.168.1.0 [110/2] via 172.16.1.2, 00:16:21, GigabitEthernet0/2
```
(##Notice how our 172.18 networks are now summarized to the 172.18.0.0 /22 route.)

Step 4: On R2, let's summarize the 172.17 networks assigned to R2's loopback interface. The 172.17 networks will be redistributed from OSPF Area 1 into OSPf Area 0.

```
R2>en
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#router ospf 1
R2(config-router)#area 1 range 172.17.0.0 255.255.252.0 ?
  advertise      Advertise this range (default)
  cost           User specified metric for this range
  not-advertise  DoNotAdvertise this range
  <cr>           <cr>

R2(config-router)#area 1 range 172.17.0.0 255.255.252.0
R2(config-router)#end
```

(##We are always going to put the area that we are summarizing from when we are summarizing our networks.)
(##Notice how we can specify the cost of a route as we are injecting it into another area. If we don't specify the cost of the route, Cisco IOS is going to choose the best metric, or the lowest cost of all the routes to assign to the summarized route.)

Step 5: Let's go back to R1 and see if we can now see the summarized network from R2 in R1's routing table.

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static
route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
C        10.1.1.0/24 is directly connected, GigabitEthernet0/1
L        10.1.1.1/32 is directly connected, GigabitEthernet0/1
```

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```
O E2     10.2.2.0/24 [110/20] via 172.16.1.2, 00:42:34, GigabitEthernet0/2
      172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C        172.16.1.0/30 is directly connected, GigabitEthernet0/2
L        172.16.1.1/32 is directly connected, GigabitEthernet0/2
      172.17.0.0/22 is subnetted, 1 subnets
O IA     172.17.0.0 [110/2] via 172.16.1.2, 00:19:25, GigabitEthernet0/2
      172.18.0.0/22 is subnetted, 1 subnets
O E2     172.18.0.0 [110/20] via 172.16.1.2, 00:29:04, GigabitEthernet0/2
      192.168.1.0/30 is subnetted, 1 subnets
O IA     192.168.1.0 [110/2] via 172.16.1.2, 00:42:56, GigabitEthernet0/2
```

(##Notice how our 172.18 networks are now summarized to the 172.18.0.0 /22 route.)

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